

# Systems Under Pressure

Shared Group Investigation Notebook

DRIVING QUESTION

How do we know when a system is actually in danger?

GROUP DETAILS

GROUP NAME \_\_\_\_\_

MEMBERS \_\_\_\_\_  
\_\_\_\_\_

INVESTIGATION TOPIC \_\_\_\_\_

CATEGORY \_\_\_\_\_

ESSENTIAL QUESTION \_\_\_\_\_

HOW THIS NOTEBOOK WORKS

- One shared investigation journal per group.
- Individual handwritten extensions are separate and shown during vlogs.
- Three informal vlogs: sequences, exponentials, polynomials.
- Short handwritten reflections live inside the individual extensions.

CORE ROUTINES

01 SYSTEM -> VARIABLES -> MODEL -> CRITIQUE

02 THINK FIRST -> SPARKY -> VERIFY -> DECIDE

03 LOCAL CHECK vs GLOBAL CHECK

04 WHAT THE MODEL REVEALS / HIDES

05 ONE REJECTED MODEL

THE WORK OF THIS NOTEBOOK

Graphs, equations, calculations, annotations, revision, evidence, and one rejected model.

Before building any model, understand the situation: who is affected, what they experience, what the data shows, and what it hides.

MAP

Use this page for free exploration: what do we know, what surprises us, what makes this urgent, and what investigation question should we draft?

Use this page to identify who is affected, what they experience, what variables they would track, and where perspectives conflict.

| SYSTEM                                                       | VARIABLES                                                                     | MODEL                                                          | CRITIQUE |
|--------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------|----------|
| <div>Stakeholders</div> <div>Who is affected, and how?</div> | <div>Variables</div> <div>What would each person measure or care about?</div> | <div>Conflict</div> <div>Where do perspectives disagree?</div> |          |
| OPEN WORKSPACE                                               |                                                                               |                                                                |          |

Use this page to map connected variables, possible hidden pressures, measurable evidence, and at least one confounding variable.

## OPEN WORKSPACE

# Graph Observations

Use this page to annotate trends, thresholds, anomalies, and intervals before choosing equations.

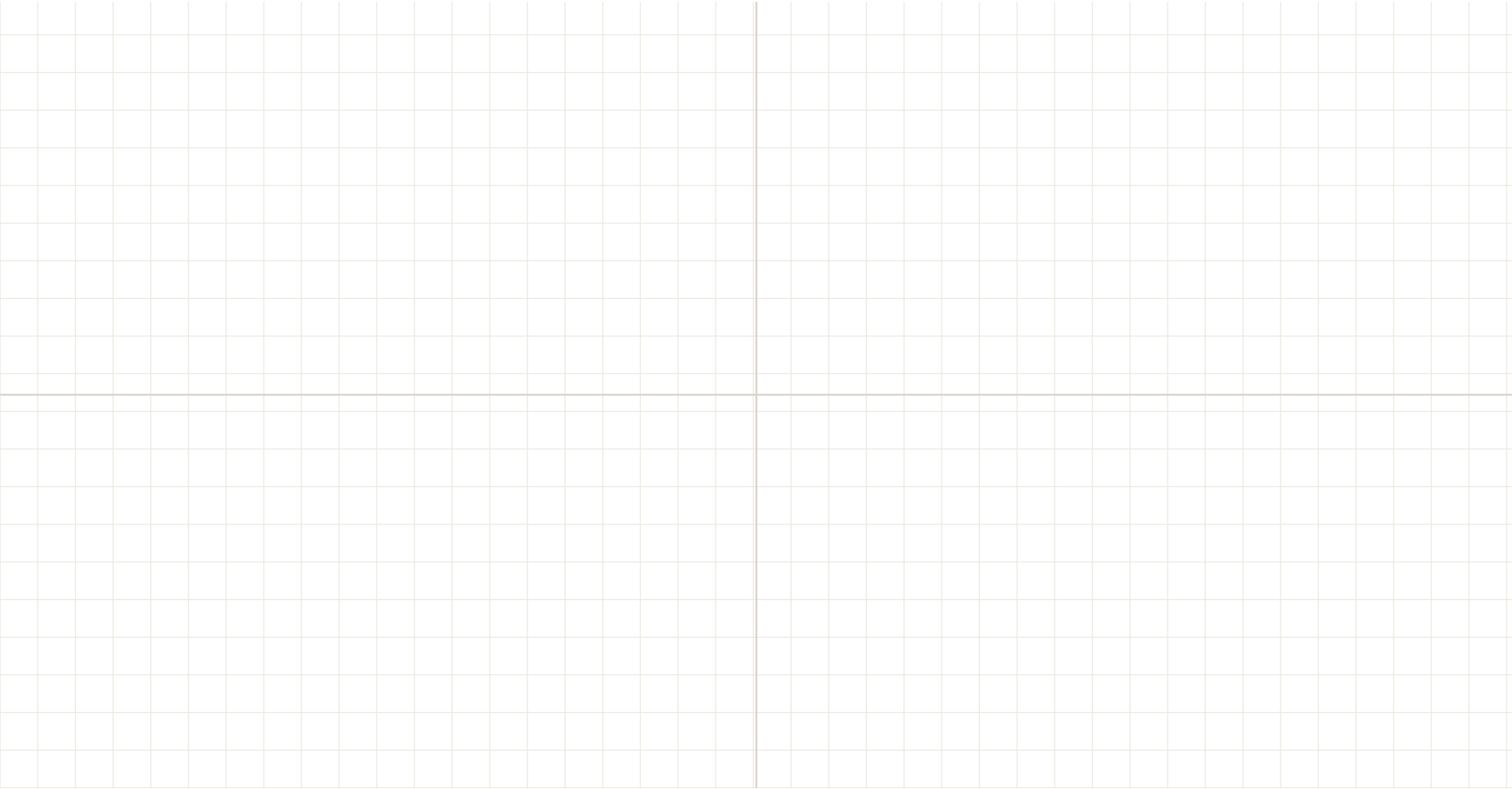
| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

Graph First  
What feature matters most?

Interval  
Where does the pattern change?

Evidence  
Which part supports your claim?

OPEN WORKSPACE



# Local vs Global Behavior

Use this page to compare a short interval with a longer interval and notice model failure.

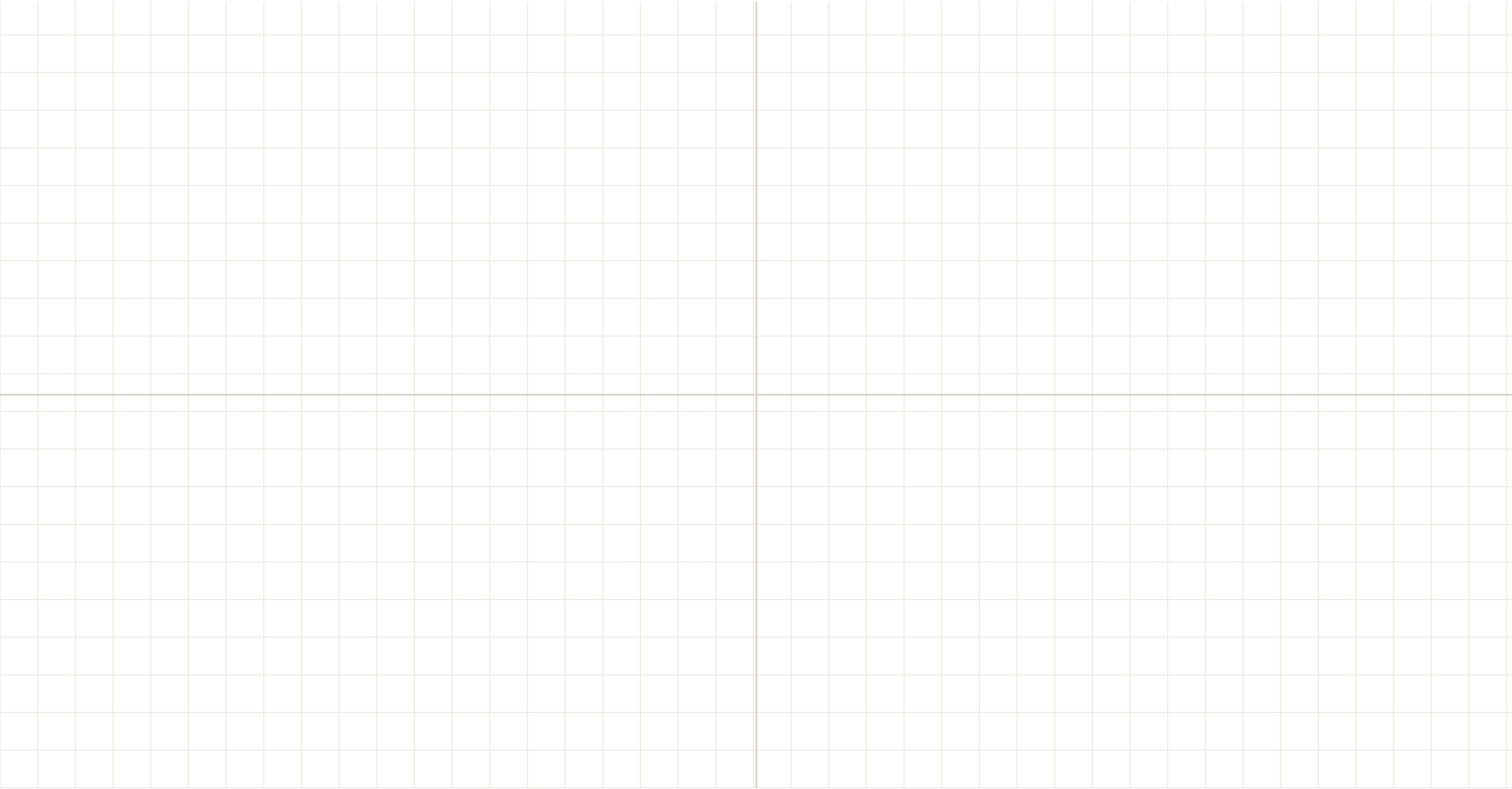
| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

Zoom In  
What seems true here?

Zoom Out  
Does it still make sense globally?

Bad Extrapolation  
Where did the model stop being meaningful?

OPEN WORKSPACE



# Sequences + Series Modeling

Use this page to model repeated change, accumulation, prediction, and where the pattern breaks.

| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

**Recursive**  
How does each term depend on the one before it?

recursive                       $a_n = r \cdot a_{n-1} + d$

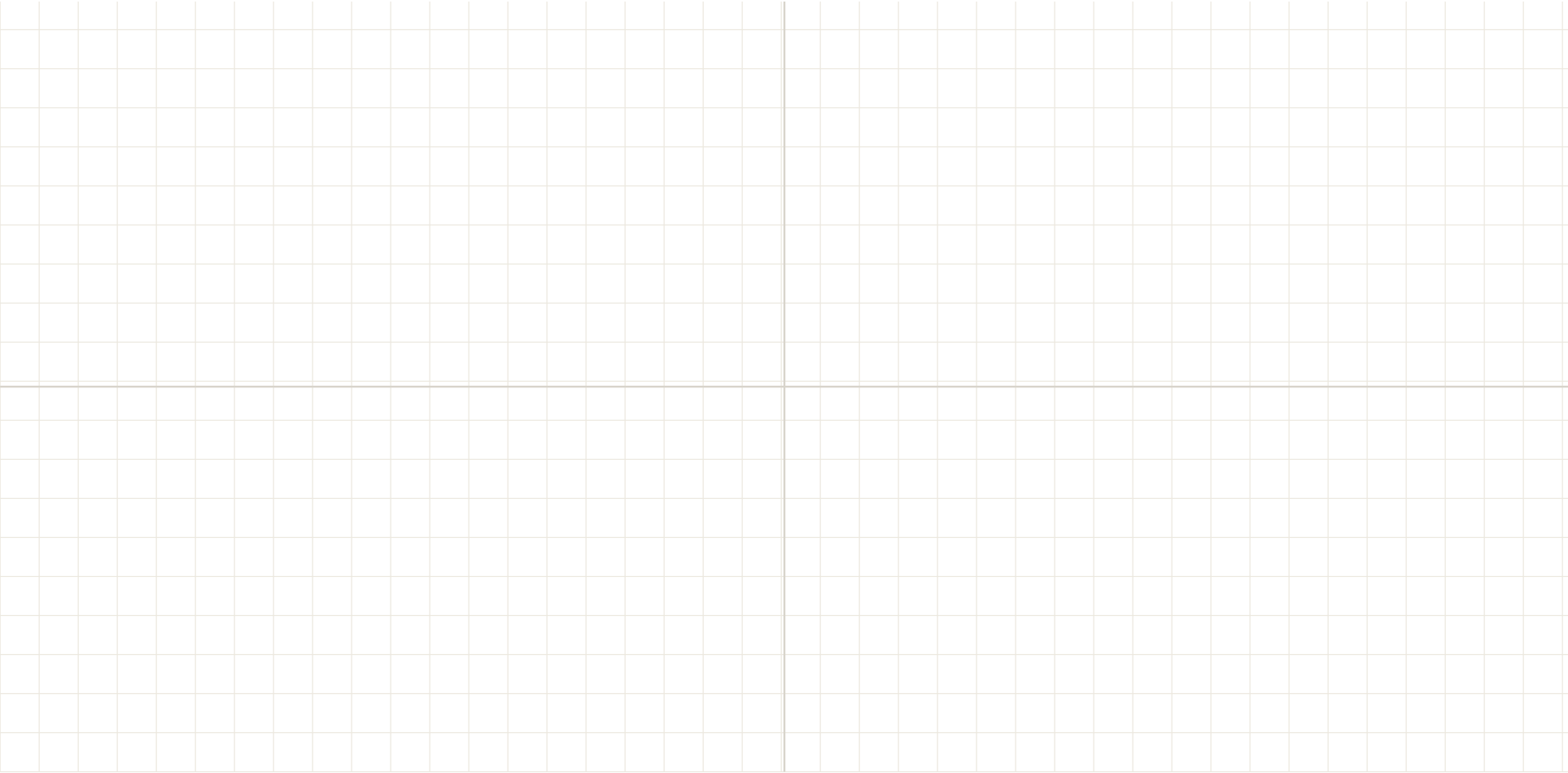
**Explicit**  
Can you predict term n directly?

explicit                       $a_n = a_1 r^{n-1}$

**Accumulation**  
What does the partial sum represent?

series                       $S_n = a_1 + a_2 + \dots + a_n$

OPEN WORKSPACE



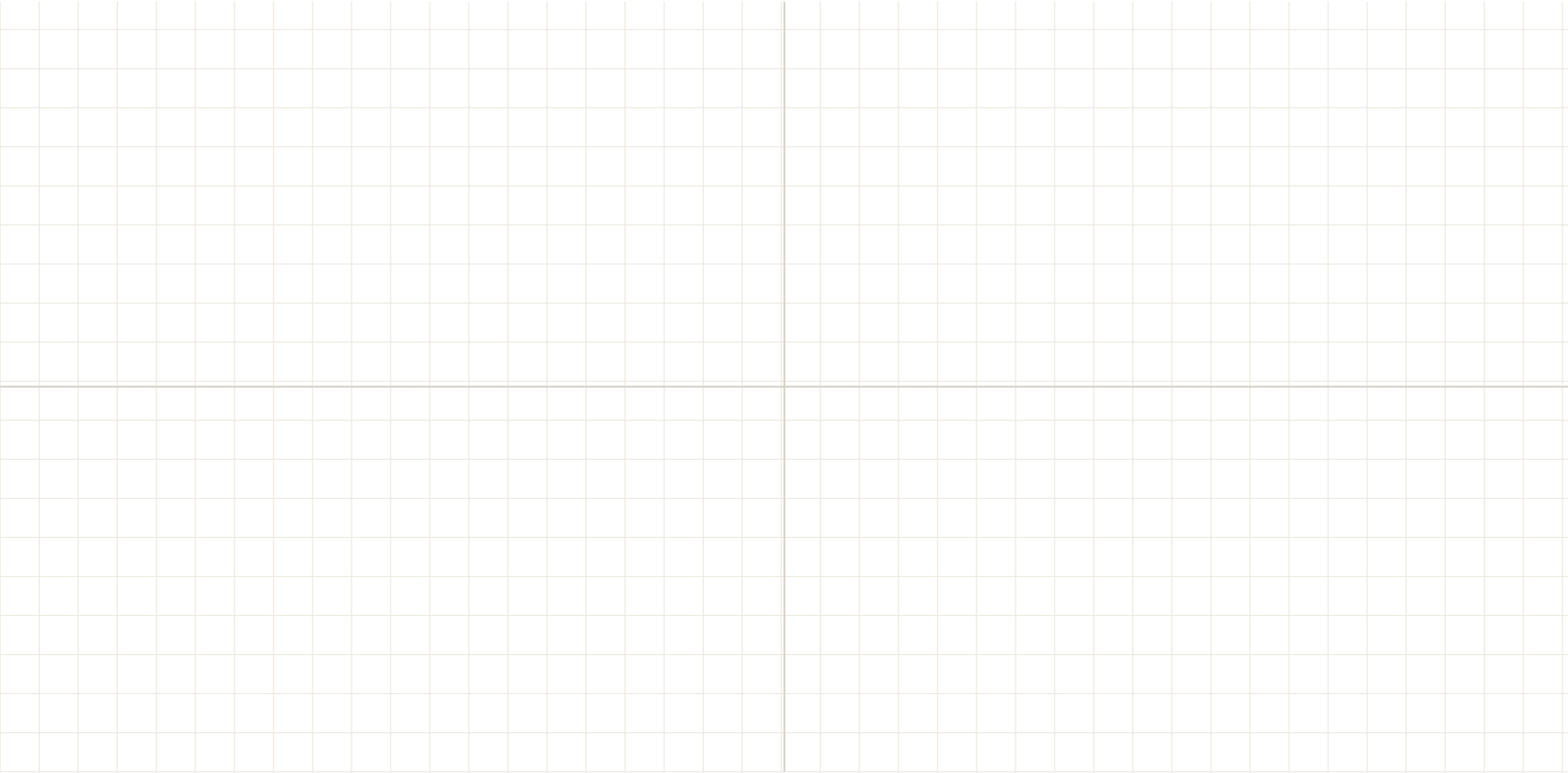
# Exponentials + Thresholds

Use this page to test ratios, solve thresholds, interpret parameters, and name where exponential reasoning fails.

| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

|                                                                                                                            |                                                                                                                                           |                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Ratio Test</div> <div>Are the ratios stable enough?</div> <div>model</div> <div><math>f(x) = a \cdot b^x</math></div> | <div>Threshold</div> <div>What value matters in the system?</div> <div>solve</div> <div><math>\log_b(\text{threshold}/a) = x</math></div> | <div>Failure</div> <div>Where does growth/decay become unrealistic?</div> <div>meaning</div> <div>a = initial value, b = growth/decay factor</div> |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|

OPEN WORKSPACE



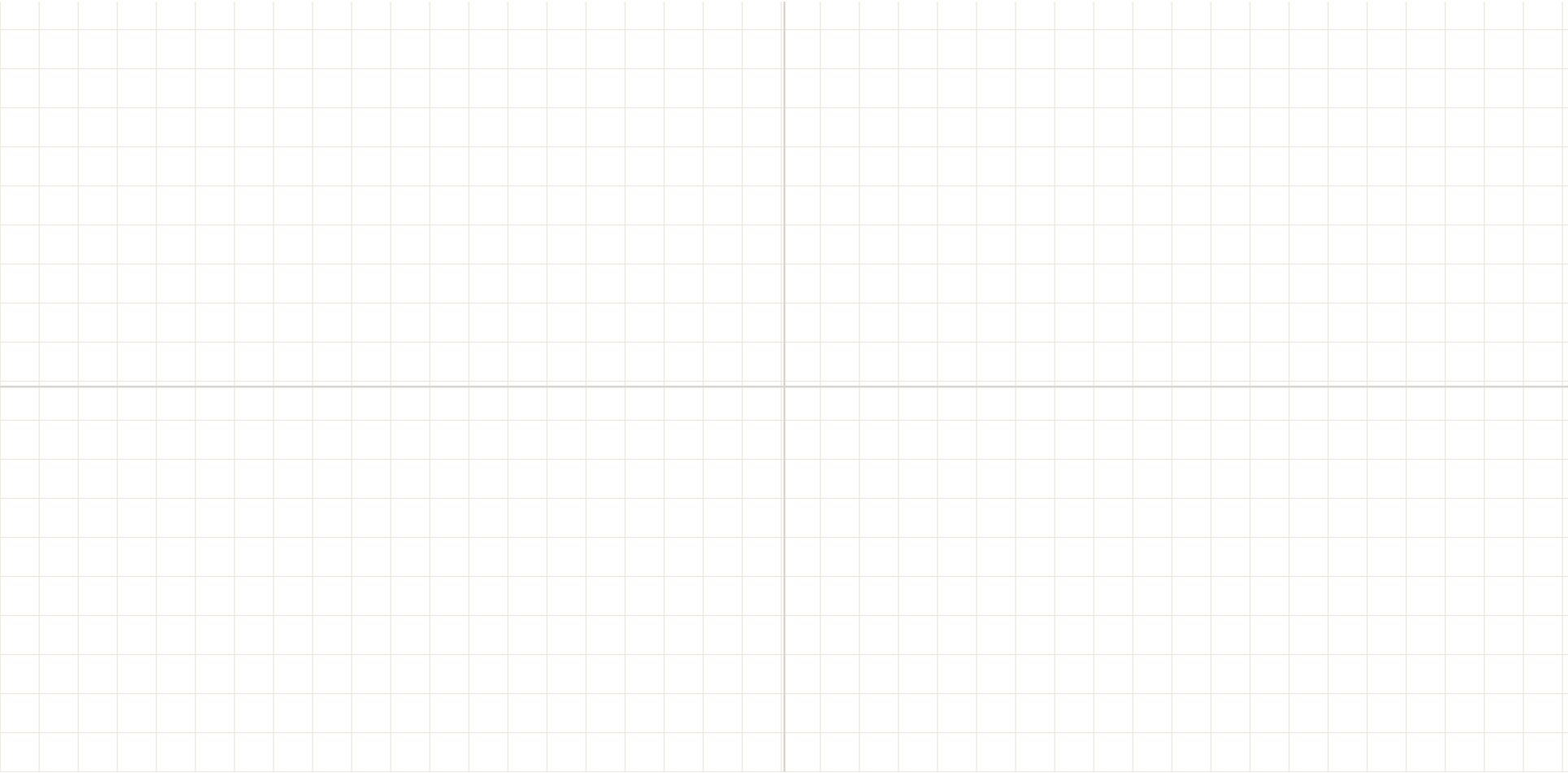
# Polynomial Behavior

Use this page to interpret local shape, turning points, zeros, thresholds, multiplicity, and end behavior realism.

| SYSTEM | VARIABLES | MODEL | CRITIQUE |
|--------|-----------|-------|----------|
|--------|-----------|-------|----------|

|                                                                                                                                               |                                                                                                                            |                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Local Model</div> <div>What interval justifies this shape?</div> <div>factored</div> <div><math>P(t) = a(t-r_1)(t-r_2)^2...</math></div> | <div>Turning Point</div> <div>What does the reversal mean physically?</div> <div>zero</div> <div><math>P(t)=0</math></div> | <div>End Behavior</div> <div>Where does the polynomial stop making sense?</div> <div>feature</div> <div>multiplicity: cross or bounce?</div> |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

OPEN WORKSPACE



Use this page to compare models by what they predict, reveal, hide, assume, and fail to explain.

## OPEN WORKSPACE

## Confounding Variables

Use this page to separate visible correlation from possible causes and alternative explanations.

| SYSTEM                                                           | VARIABLES                                                   | MODEL                                                                                                                             | CRITIQUE |
|------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| <div>Related</div> <div>What evidence suggests connection?</div> | <div>Not Proof</div> <div>What else could explain it?</div> | <div>Sentence Frame</div> <div>These variables appear related because ____, but this does not prove causation because ____.</div> |          |
| OPEN WORKSPACE                                                   |                                                             |                                                                                                                                   |          |

Use this page only after your group thinks first: record what Sparky suggested, what you verified, and what you rejected.

## OPEN WORKSPACE

Use this page to show how evidence, critique, or model failure changed your group's interpretation.

## OPEN WORKSPACE

Use this page to make a defensible claim with graph evidence, calculation evidence, rejected-model evidence, and uncertainty.

| SYSTEM                                                      | VARIABLES                                                        | MODEL                                                                             | CRITIQUE |
|-------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------|
| <div>Graph</div> <div>Which graph supports the claim?</div> | <div>Calculation</div> <div>Which calculation supports it?</div> | <div>Counterargument</div> <div>What evidence would change your conclusion?</div> |          |
| OPEN WORKSPACE                                              |                                                                  |                                                                                   |          |

Use this page to prepare a mathematical argument: strongest evidence, biggest limitation, rejected model, and final defense.

| Explain                            | Defend                                                              | Close                                   |
|------------------------------------|---------------------------------------------------------------------|-----------------------------------------|
| Every graph and equation verbally. | Mathematical, realism, assumption, interval, counterargument tests. | Not a summary. A mathematical argument. |

## OPEN WORKSPACE

Use this page for last annotations, slide notes, and questions your group must be ready to answer.

|                                                   |                                                                         |                                                                       |
|---------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Ready</b><br>All members can explain the math. | <b>Artifacts</b><br>Notebook, vlogs, extensions, and sources are ready. | <b>Attend</b><br>Presentation day is during class on Tuesday, June 9. |
|---------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|